

Microcontrollers Milestone 1

Line following RC car

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1. Project Handling Strategy

To ensure the robot operates reliably under varying conditions, the project will follow a modular development approach. This separates the hardware abstraction (sensor reading and motor driving) from the high-level logic.

- Phase A: Hardware Integration & Calibration: Focus on normalizing IR sensor values to handle different lighting and surface reflectivities.
- Phase B: Control Logic Development: Implementing a PID (Proportional-Integral-Derivative) Lite approach for smooth turning
- Phase C: State Management: Developing the Finite State Machine (FSM) to handle transitions between active tracking, line loss, and manual override (start/stop switch).

2. Constraints & Priorities

- Priority 1: Reliability of Detection: The car must accurately distinguish the line from the floor across different ambient light levels.
- Priority 2: Smooth Motion: Preventing shaking is critical. This is achieved via PWM (Pulse Width Modulation) to slow down the inner wheel during turns rather than stopping it.
- Constraint 1: Power Management: the system must use a dedicated power source (, Li-ion battery) to prevent the microcontroller from resetting during motor spikes.
- Constraint 2: Sensor Geometry: The physical spacing of the 5-sensor array must match the width of the black tape used for the track.

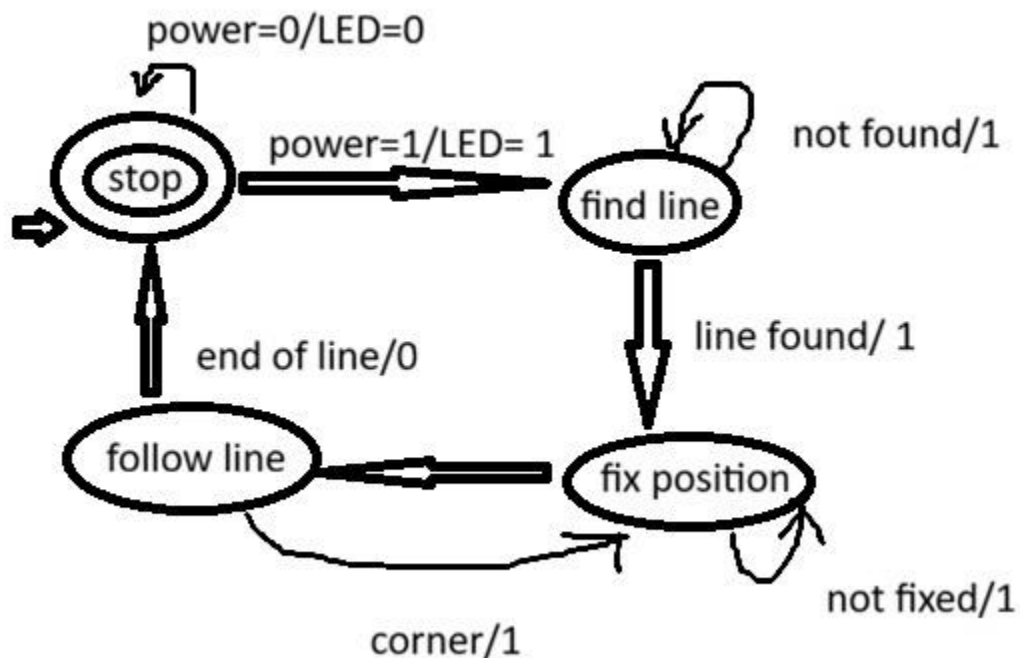
3. Full System Description

The system is an autonomous mobile platform powered by an ATmega32.

It uses an Infrared (IR) Reflectance Array to detect the contrast between a dark line and a light surface.

- Sensing: The IR sensors output analog or digital signals based on the light reflected. These are processed to calculate an "error" value (how far the car is from the center).
- Processing: The controller calculates the necessary motor speeds using PWM. If all sensors read "white," the system triggers a Recovery State.
- Actuation: An L298N H-Bridge motor driver manages two DC motors, allowing for independent speed and direction control.
- User Interface: A physical toggle switch for safety, while a high-visibility LED provides immediate feedback on the system's operational status.

4. Finite State Machine



5. Bill Of Materials

Component Name	Quantity	Price
Control / Processing		
Atmega32	1	230
Crystal 10MHz	1	4
AVR programmer	1	175
Actuation		
L298N Driver	1	75
Geared motor	2	-
Servo Motor	1	85
Sensing		
IR sensors	5	175
Mechanical		
65mm wheel	2	162
Caster Wheel	1	30
Plywood board	1	92
Power		
Li ion batteries	3	141
Battery Pack AAx4	1	22
Resistors/Capacitors	17	~5
Wiring		
Bread board	1	75
Jumper wires	30	45
Total Price		1316.5